

Common network architectures

COMP6252 (Deep Learning Technologies)

ECS, University of Southampton

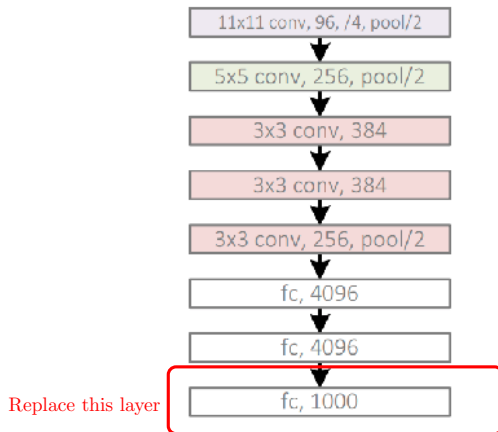
Transfer learning

- Basically transfer learning is using pre-trained models
- The model was already trained on a **different** datasets
- The idea is the pre-trained network has learned "high level" features common to all images
- What is needed is some fine tuning
- This is typically done by replacing the last layer of the pre-trained model
- We will use the AlexNet model for illustration because it is not too big.
- The idea is the same for all pre-trained networks

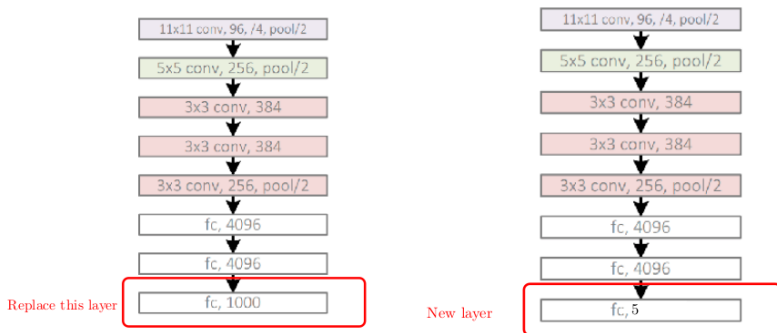
Replacing the last layer

- Almost all available (vision) models were trained on ImageNet competition
- The last layer outputs 1000 values
- We will use a model pre-trained on ImageNet to classify flowers (5 types)

Transfer learning

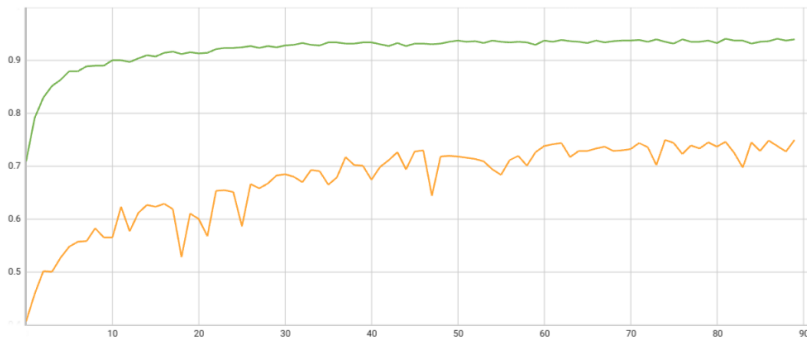


Transfer learning



Why not start from scratch?

- Validation accuracy vs epoch
- ResNet18 pre-trained (all layers): green
- ResNet18 trained with random initial weights: orange



How to train a pre-trained model

- There are three approaches for training pre-trained models on a different dataset
 - 1 "freeze" the weights of all intermediate layers and train the last layer
 - 2 train all the layers
 - 3 A hybrid approach where only the last layer is trained first followed by "unfreezing" the intermediate layers
- The accuracy of different approaches is dataset dependent

Last vs all

■ Validation accuracy vs epoch

- 1 All layers are trained: orange
- 2 Only the last layer is trained (intermediate ones are frozen): green

